

# How to Use the Class File for the IEICE Transactions\*\*

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**SUMMARY** IEICE (The Institute of Electronics, Information and Communication Engineers) provides a  $\text{\LaTeX} 2_{\epsilon}$  class file, named `ieice.cls`, for the IEICE Transactions. This document describes how to use the class file, and also makes some remarks about typesetting a manuscript by using the  $\text{\LaTeX} 2_{\epsilon}$ . The design is based on  $\text{\LaTeX} 2_{\epsilon}$ .

**key words:**  $\text{\LaTeX} 2_{\epsilon}$  class file, typesetting, math formulas

## 1. Introduction

This document describes how to handle the `ieice.cls` for the IEICE (the Institute of Electronics, Information and Communication Engineers) Transactions. Section 2 explains how to typeset according to the template. `template.tex` which is distributed with the `ieice.cls` can be used. Section 3 describes a special feature of `ieice.cls`, which is different from the `article.cls` provided by the standard  $\text{\LaTeX} 2_{\epsilon}$  and which points may be aware of on writing a manuscript and so on. Section 4 is about typographic notes, which explains how to typeset, how to prevent typographic errors and how to handle long formulas. Section 5 describes some notations when submitting of editable electronic files for publishing. For information about printing on A4 paper and making pdf file, see Appendix A.

## 2. Template and How to Typeset a Manuscript

`ieice.cls` should be specified as a document class, not as an option. The layout is influenced with the following formatting request, `twocolumn`, `twoside` and `fleqn`, which are declared inside the class file. There is no need to specify them as an option again and other options changing the layout or all style parameters should not be specified.

### 2.1 The Class Option

Information for Authors (Brief Summary) says that each submitted manuscript is categorized as either “PAPER” or “LETTER”. `ieice.cls` provides options of `\documentclass` for not only “PAPER” and “LETTER” but also “INVITED PAPER”, “SURVEY PAPER”, and so on shown in Table 1.

**Table 1** Options of `\documentclass`.

| class option               | manuscript type      |
|----------------------------|----------------------|
| <code>paper</code>         | PAPER                |
| <code>invited</code>       | INVITED PAPER        |
| <code>position</code>      | POSITION PAPER       |
| <code>survey</code>        | SURVEY PAPER         |
| <code>invitedsurvey</code> | INVITED SURVEY PAPER |
| <code>review</code>        | REVIEW PAPER         |
| <code>tutorial</code>      | TUTORIAL PAPER       |
| <code>letter</code>        | LETTER               |
| <code>brief</code>         | BRIEF PAPER          |

With no optional arguments to `\documentclass`, `ieice.cls` will be formatted in “PAPER” style.

When `letter` or `brief` (only for Trans. on Electronics) is specified, the author’s profile (the `\profile` command, see page 3) at the end of a manuscript will not be produced.

### 2.2 Template

Here is the template.

```
\documentclass[paper]{ieice}
%\documentclass[invited]{ieice}
%\documentclass[position]{ieice}
%\documentclass[survey]{ieice}
%\documentclass[invitedsurvey]{ieice}
%\documentclass[review]{ieice}
%\documentclass[tutorial]{ieice}
%\documentclass[letter]{ieice}
%\documentclass[brief]{ieice}
%\usepackage[dvips]{graphicx}
%\usepackage[pdftex]{graphicx,xcolor}
\usepackage[dvipdfmx]{graphicx,xcolor}
\usepackage[fleqn]{amsmath}
\usepackage{newtxtext}
\usepackage{varg}{newtxmath}
\setcounter{page}{1}
\field{A}
%\vol{107}
%\no{1}
\SpecialSection{\LaTeX\ Class File
                for the IEICE Transactions}
%\SpecialIssue{}
\title[How to Use the Class File]
      {How to Use the Class File
       for the IEICE Transactions}
\titlenote{This paper was presented at ...}
```

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Manuscript revised April 1, 2024.

<sup>†</sup>The Faculty ...

<sup>††</sup>The Faculty ...

\*Presently,

\*\*This paper was presented at ...

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```

\authorlist{%
  \authorentry{Hanako DENSHI}{m}{labelA}
  \MembershipNumber{1111111}
  \authorentry{Taro DENSHI}{n}{labelB}{labelC}
  \MembershipNumber{}
}
%\breakauthorline{2}
\affiliate[labelA]
{The Faculty ...}
\affiliate[labelB]
{The Faculty ...}
\paffiliate[labelC]
{Presently, }
\received{2015}{10}{8}
\revised{2024}{4}{1}

\begin{document}
\maketitle

\begin{summary}
IEICE (The Institute of Electronics,
Information and Communication Engineers)
provides a \LaTeXe\ class file,
named \ClassFile, for the IEICE Transactions.
...
\end{summary}
\begin{keywords}
\LaTeXe\ class file, typesetting,
math formulas
\end{keywords}
\section{Introduction}
... ..
\section*{Acknowledgments}
... ..
\bibliographystyle{ieicetr}
\bibliography{myrefs}
\begin{thebibliography}{9}
\bibitem{}
\end{thebibliography}

\appendix
%\appendix*
... ..
\profile{Hanako Denshi}{received the B.S. and
M.S. degrees in Electrical Engineering from
Denshi Institute of Technology in 1997
and 1999, respectively. ...}
\end{document}

```

- The `\field` command is required by the header. Its argument indicates the categories of Transactions (see following table). For example, in the case of “Fundamentals”, A is specified as an argument of `\field`.

|   |                         |
|---|-------------------------|
| A | Fundamentals            |
| B | Communications          |
| C | Electronics             |
| D | Information and Systems |

- The `\vol` and `\no` commands are not needed to be assigned in submitted manuscript (there are no entries of them in `template.tex`). They are required by the header, which needs information of volume and number of issue in final printing, and are assigned as `\vol{107}` and `\no{1}` respectively for example.
- The `\SpecialSection` and `\SpecialIssue` commands are required in the case of submission to the Special Section (Special Issue) described on Call for Papers, not in the case of the Regular Section. An argument description is as follows.

```
\SpecialSection{Image Processing}
```

- The title of a manuscript is assigned in `\title`. You may use `\\` to start a new line in a long title. The argument of the `\title` command is required for more than just producing a title, it is also required to generate a running head, combining with authors' names. If you want a shorter title for a running head, type as follows.

```
\title[short title]{title}
```

- If you need to describe a notation when a manuscript was first reported and by which organization authors were supported, etc., the `\titlenote` command can be used.
- The outputs of authors' names, membership status and marks of affiliates are automatically generated by using the `\authorlist` and `\authorentry` commands. The `\authorentry` command must be described as an argument of the `\authorlist` command. The `\authorentry` command has three arguments.

```

\authorentry{name}{membership}{label}
\MembershipNumber{membership number}

```

For example, they could be typed as follows.

```

\authorlist{%
  \authorentry{Hanako DENSHI}{m}{labelA}
  \MembershipNumber{1111111}
  \authorentry{Taro DENSHI}{n}{labelB}
  \authorentry{Min~Jun Kim}{n}{labelC}
}

```

- The first argument of `\authorentry` is filled with an author's name. The family name might be described in uppercase letters. This command automatically capitalizes a word or words after a first space in the first argument. Combine a first name and a middle name with “~” to handle a middle name rightly.
- The second argument is specified by one letter out of five letters (m, n, a, s, h, f, e), each one indicating the membership status of each author shown in Table 2.

To specify other letters will not cause errors, but will cause wrong output. No extra spaces may be added between a letter and a brace. `{m}` and `{m_}` are regarded as different. The latter will not

**Table 2** Membership Status

|   |                         |
|---|-------------------------|
| m | Member                  |
| n | Nonmember               |
| a | Affiliate Member        |
| s | Student Member          |
| h | Fellow, Honorary Member |
| f | Fellow                  |
| e | Senior Member           |

the left column is letters to be specified. the right column is membership status to be generated.

generate “Regular Member”.

- The third argument is assigned by the label of the author’s affiliate, corresponding to the label of the `\affiliate` command (see below). For example, an abbreviation for a university, institute or company is recommended for the label.

In the case of no affiliate, the label `none` must be specified. And in the case of plural affiliates, labels should be specified as a comma separated list.

- Author’s membership number of IEICE must be specified by using `\MembershipNumber`. In the case of nonmember, `\MembershipNumber` must be left its argument blank.
- E-mail addresses might be specified. Its description is as follows.

```
\authorentry[name@xx.yy.zz]
{Hanako DENSHI}{m}{labelA}
```

- If you need to inform a present affiliate, the optional fourth argument of `\authorentry` can be used as follows.

```
\authorentry{Hanako DENSHI}{m}{labelA}
[labelB]
```

The fourth argument which is described in brackets is corresponding to the label of the `\paffiliate` command (see below).

- The `\breakauthorline` command is provided, if you would like to break a line of author’s lists at any point. `\breakauthorline{num,num,num,...}` *num* must be a positive integer. If “3” is specified, the line-break will be occurred after the third author. If “2,4,6” is specified, line-breaks will be occurred after the second, fourth and sixth authors.
- Author’s affiliate is described in the `\affiliate` command as follows.

```
\affiliate[label]{affiliate}
```

The first argument `label` must be the same as the 3rd argument of the `\authorentry` command. No extra spaces may be added between a letter and a brace. The second argument is filled with the author’s affiliate.

The entry of `\affiliate` must be put in the same order as labels of `\authorentry` lists.

If the labels of `\affiliate` are different from those of `\authorentry`, there will come a warning message on

your terminal.

- The author’s present affiliate is described in the `\paffiliate` command as follows.

```
\paffiliate[label]{present affiliate}
```

The first argument must be the same as the fourth argument of `\authorentry` command.

- The `\received` and `\revised` commands are required to generate the date of receipt of a manuscript, revision of a manuscript and the date of final version. Those descriptions is in the order of year/month/day. For example, the date of receipt is assigned as `\received{2015}{10}{8}`.

All those commands should be written in preamble.

The `\maketitle` command should be placed after the `\begin{document}` command. It generates the title.

- The text of the abstract is described in the `summary` environment. It should be about 300 words for a “PAPER”, 50 for a “LETTER” in a single paragraph.
- The text of the keywords is described in the `keywords` environment. The text should be 4–5 words and be given in lowercase letters except abbreviations and proper nouns.
- If you might express your gratitude, the following description is recommended.

```
\section*{Acknowledgments}
```

- The `\appendix` command provided by the standard  $\LaTeX 2_{\epsilon}$  is only a declaration that changes the way sectional units are numbered. But `\appendix` and `\appendix*` commands provided by `ieice.cls` are different from it.

Once the `\appendix` command is declared, the following `\section` commands will generate “Appendix A:”, “Appendix B:”, ... . On the other hand, the `\appendix*` command will generate “Appendix:” without sectional numbers. So the latter should be used when the appendix has no more than one section.

Once either of both commands is declared, equation numbers and float numbers are numbered “A·1”, “A·2”, ... .

- Authors’ biographies (not necessary for a “LETTER”) on page 8 are generated with:

```
\profile{Hanako Denshi}
{received the B.S. and M.S. degrees
...}
```

- The first and second arguments are filled with an author’s name and profile respectively.
- If EPS or PDF (see page 4) files of pictures of the authors’ faces are provided, put the EPS (or PDF) files named `a1.eps`, `a2.eps` (or `a1.pdf`, `a2.pdf`), etc., which are followed by the order of authors, on the current directory of your computer. The `\profile` command automatically reads their files and puts their pictures on the left margin.

You might specify a eps (or pdf) file name as follows.

```
\profile[file.eps]{Hanako Denshi}{...}
\profile[file.pdf]{Hanako Denshi}{...}
```

When compiling by pdflatex, you might specify a pdf file name as follows,

```
\profile[file.pdf]{Hanako Denshi}{...}
```

- graphics or graphicx package must be specified.
- The ratio of EPS (or PDF) file must be width : height = 25 : 33. EPS (or PDF) files will be read by the following command.

```
\resizebox{25mm}{!}
{\includegraphics{a1.eps}}
```

- If their files don't exist in the current directory, simple frames will be generated (see page 8).

Pictures of the authors' faces may be omitted by using the `\profile*` command instead of the `\profile` command.

### 3. Special Feature of ieice.cls and Notes about Some Features of L<sup>A</sup>T<sub>E</sub>X 2<sub>ε</sub>

#### 3.1 Formula

As described in Sect. 2, the `fleqn` option is in effect. A displayed formula is aligned on the left, a fixed distance (7 mm) from the left margin, instead of being centered. A formula number is put on the right side.

Although a width of one column might be felt too narrow to compose displayed formulas, equations should be composed with the proper length, paying attention to the message “Overfull \hbox”. Section 4.2 describes several solutions and hints to handle a long formula.

#### 3.2 Figures and Tables

The font size inside the `figure` and `table` environments is set `\footnotesize` (8 pt) (see Table 3).

The `[h]` option, one of the arguments of floating environment specifying a location where the float may be placed, is not recommended. Figures and tables should be located at the top or bottom of a page by using `[t]`, `[b]` or `[p]`.

##### 3.2.1 Including Graphics

Although there are many ways to include pictures and figures in L<sup>A</sup>T<sub>E</sub>X, the Encapsulated POSTSCRIPT format (EPS) or PDF is recommended.

Here is a simple explanation to insert graphics into a manuscript.

The `graphicx` package must be loaded. The option `dvips`, `pdftex` or `dvipdfmx` is one of the device driver's

**Table 3** The font size in the `table` environment is 8 point.

| A | B | C |
|---|---|---|
| X | Y | Z |

```
\begin{table}[tb]
\caption{An example of table.}
\label{table:1}
\begin{center}
\begin{tabular}{c|c|c}
\hline
A & B & C\\
\hline
X & Y & Z\\
\hline
\end{tabular}
\end{center}
\end{table}
```

option, it might be changed according to a device driver you use.

```
%% for eps
\usepackage[dvips]{graphicx}
%% for pdflatex
\usepackage[pdftex]{graphicx,xcolor}
%% for platex or uplatex
\usepackage[dvipdfmx]{graphicx,xcolor}
```

A graphics file (for example, EPS file) can be included with the `\includegraphics` command.

```
\begin{figure}[tb]
\begin{center}
\includegraphics{file.eps}
\end{center}
\caption{...}
\label{fig:1}
\end{figure}
```

If the option `scale=0.5` is given, the graphics will be scaled by half.

```
\includegraphics[scale=0.5]{file.eps}
```

You can get the same result as above by using the `\scalebox` command.

```
\scalebox{0.5}{\includegraphics{file.eps}}
```

If the option `width=30mm` is given, the width of graphics will be 30 mm (with the height proportionally scaled).

```
\includegraphics[width=30mm]{file.eps}
```

The next is another example using `\resizebox`.

```
\resizebox{30mm}{!}
{\includegraphics{file.eps}}
```

Both dimension of width and height can be specified as follows.

```
\includegraphics[width=30mm,height=40mm]
{file.eps}
```

or

```
\begin{figure}[tb]
... floating materials ...
\capwidth=50mm
\caption{An example of figure.}
\label{fig:1}
\end{figure}
```

Fig. 1 An example of figure.

```
\resizebox{30mm}{40mm}
{\includegraphics{file.eps}}
```

For further information about the graphics package, see reference book [7], [9].

### 3.2.2 Captions of Floating Environment

ieice.cls sets the width of caption to about 83.5 mm ( $\text{\columnwidth}$ ) in the case of single column and about 116 mm ( $0.66\text{\textwidth}$ ) in the case of double column.

The width of caption can be set by changing the value of `\capwidth` (see Fig. 1).

### 3.3 Theorem-like Environment

If you use the `\newtheorem` environment, pay attention to the following points. Additional vertical spaces before and after the environment are `.5\baselineskip`, and the text within the environment does not appear in italics.

An example is given as follows.

```
\newtheorem{theorem}{Theorem}
\begin{theorem}[Fermat]
There are no positive integers such that
 $x^n + y^n = z^n$  for  $n > 2$ .
I've found a remarkable proof of this fact,
but there is not enough space
in the margin [of the book] to write it.
(Fermat's last theorem).
\end{theorem}
```

**Theorem 1** (Fermat): There are no positive integers such that  $x^n + y^n = z^n$  for  $n > 2$ . I've found a remarkable proof of this fact, but there is not enough space in the margin [of the book] to write it. (Fermat's last theorem).

### 3.4 Footnotes

The footnote begins with “†” (see page 7). As the footnote counter increases, the footnote marks proceed as “†”, “††”, “†††”. The footnote mark is set to reset at each page.

### 3.5 Bibliography and Citations

The bibliographic reference list should be generated according to the IEICE editing style, e.g., authors' initials, names, title of article, journal abbreviation, volume, number, pages, and publication year, etc. Information about composing such

lists can be given in “Information for Authors (Brief Summary)” and the following web site.

<https://www.ieice.org/eng/shiori/index.html>

On the other hand, in case using Bib<sub>T</sub><sub>E</sub>X [5] the bibliography style ieicetr.bst (numeric citation order) is recommended, which is distributed with ieice.cls.

ieice.cls includes the citesort package with a slight modification. The citesort package collapses a list of three or more consecutive numbers into a range, and sorts the numbers before collapsing them. For instance, while the following example, `\cite{FMi1,FMi2,FMi3,latexbook,texbook,Salomon}`, would produce [5, 9, 10, 7, 1, 8] in the standard style, it is transformed into “[1], [5], [7]–[10]” in this class file.

### 3.6 Verbatim Environment

You can change the values of the parameters in the verbatim environment which is customized in ieice.cls. The default settings are:

```
\verbatimleftmargin=0pt
\def\verbatimsize{\normalsize}
\verbatimbaselineskip=\baselineskip
```

For example, those parameters can be changed as follows.

```
\verbatimleftmargin=7mm
\def\verbatimsize{\footnotesize}
\verbatimbaselineskip=3mm
```

### 3.7 AMS Packages

The  $\mathcal{AMS}$ -L<sub>A</sub>T<sub>E</sub>X packages are provided to typeset complex equations or other mathematical constructions. If you would like to use them, the amsmath package should be loaded with the fleqn option.

```
\usepackage[fleqn]{amsmath}
```

While the amsmath package presents many functions, the `\boldsymbol` command which is to be used for individual bold math symbols and bold Greek letters is needed, only the amsbsy package might be loaded.

```
\usepackage{amsbsy}
```

Once the amssymb package is loaded, many extra math symbols of the  $\mathcal{AMS}$ -L<sub>A</sub>T<sub>E</sub>X fonts will become available.

```
\usepackage{amssymb}
```

For further information about the  $\mathcal{AMS}$ -L<sub>A</sub>T<sub>E</sub>X package, see reference book [5].

### 3.8 Miscellaneous

The following macros are defined in ieice.cls.

- `\QED`: Produces “□” in the end of the proof and so on. You would get the same output by using

**Table 4** \RN and \FRAC.

| \RN{2} | \RN{117} | \FRAC{\\$pi\\$}{2} | \FRAC{1}{4}   |
|--------|----------|--------------------|---------------|
| II     | CXVII    | $\pi_2$            | $\frac{1}{4}$ |

\hfill \$\Box\$. But if the end of a paragraph goes to the right margin, the character  $\square$  is positioned at the start of a line. Using \QED will prevent such a case. Notice that the latexsym package is required to produce  $\square$ .

- \halflineskip and \onelineskip: Produce a vertical space, 0.5\baselineskip, 1\baselineskip respectively.
- \RN and \FRAC: Are shown in Table 4 [1].
- \ddash: Produce double “—”. Double “---” sometimes produce thin space between two “—”. \ddash will prevent such a case.

## 4. Typographic Notes

### 4.1 How to Prevent Typographic Errors

1. “The italic correction will be automatically added by the font commands with arguments but must be inserted manually using \/ when declarations are used” [5].
2. You should pay attention to a space after a period.  
“T<sub>E</sub>X simply assumes that a period ends a sentence unless it follows an uppercase letter. This works most of the time, but not always—abbreviations like ‘etc.’ being the most common exception. You tell T<sub>E</sub>X that a period doesn’t end a sentence by using a \\_ command (a \ character followed by a space or the end of a line) to make the space after the period.”  
“On the rare occasions that a sentence-ending period follows an uppercase letter, you will have to tell T<sub>E</sub>X that the period ends the sentence. You do this by preceding the period with a \@ command” [7].  
Beans (lima, etc.)\have vitamin B\@.
3. “Line breaking should be prevented at certain interword spaces. ... Trying ~ (a tilde character) produces an ordinary interword space at which T<sub>E</sub>X will never break a line” [7].  
Mr. ~Jones, Figure~\ref{fig:1}, (1)~gnats.
4. With respect to Figure, Section, Equation, when these words appear at the beginning of a sentence, they should be spelt out in full, e.g., “Figure 1 shows ...” is used. When they appear in the middle or the last of a sentence, abbreviations, e.g., “in Fig. 1”, “in Sect. 2”, “in Eq. (3)” should be used.
5. There should be no space after opening or before closing parentheses, as in (\\_word\\_).
6. There are many cases of an inappropriate application of a \\ or \newline command except in the tabular environment etc., such as two \\ commands in succession or \\ command just before a blank line. They will often cause warning messages like Underfull \hbox ..., as a result it would often prevent you from finding impor-

tant warning messages. The use of \par\noindent or \hfil\break commands is recommended and gives you the same effect without warning messages.

7. There are some cases of an inappropriate application of a \\ in descriptions such as program lists. Use of the tabbing environment or list environment is recommended.
8. There are three types of dashes that are used. The hyphen (-) is used in connecting English-language words such as ‘well-known’, and the en dash (--) is used when expressing a range such as ‘pp.298–301’. The em dash (---) is even longer—it’s used as punctuation.
9. Notice that L<sup>A</sup>T<sub>E</sub>X recognizes the hyphen and en dash in math mode as the minus sign. Use \hbox or \mbox if you would like to use the hyphen and en dash in math mode as it is. Some examples are given below.

$A^{b-c}$

$A^{b-c} \Rightarrow$  minus sign

$A^{\mathrm{b}}\mathrm{mbox{\scriptsize -}}$   
 $\mathrm{mbox{\scriptsize c}}$

$A^{b-c} \Rightarrow$  hyphen

$A^{\mathrm{b}}\mathrm{mbox{\scriptsize --}}$   
 $\mathrm{mbox{\scriptsize c}}$

$A^{b-c} \Rightarrow$  en dash

$A^{\mathrm{b-c}}$

$A^{b-c} \Rightarrow$  minus sign

10. The less-than sign “<” (<, a relation) should not be confused with “\langle” (\langle, a delimiter). The same is true for the greater-than sign “>” and “\rangle”.
11. A unary operator and a binary operator: “A + or – that begins a formula (or certain subformulas) is assumed to be a unary operator, so typing  $-\mathbf{x}$  produces  $-\mathbf{x}$  and typing  $\sum -\mathbf{x}_i$  produces  $\sum -\mathbf{x}_i$ , with no space between the “-” and “ $\mathbf{x}$ ”. If the formula is part of a larger one that is being split across lines, T<sub>E</sub>X must be told that the + or – is a binary operator. This is done by starting the formula with an invisible first term, produced by an \mbox command with a null argument” [7].

\begin{eqnarray}

y &= a + b + c + \dots + e \\

& & \mathrm{mbox{}} + f + \dots

\end{eqnarray}

12. \allowbreak may be used within long math formulas in paragraphs since T<sub>E</sub>X is reluctant to break lines there. It allow a line or page break where one could not ordinary occur. On the other hand, \\, \hfil\break an \linebreak force T<sub>E</sub>X to break a line at the point.

### 4.2 How to Handle Long Formulas

Here are some explanations how to handle long formulas, for example, overhanged equations, equations overriding the equation number, and so forth.

#### Example 1:

$$y = a + b + c + d + e + f + g + h + i + j + k + l + m \quad (1)$$

The above equation is too long, and the space between the equation and the equation number are too narrow and sometimes the equation number would moves to the right. In this case the `\!` command is useful.

“The \! acts like a backspace, removing the same space amount of space that \, adds” [7].

$$y = a + b + c + \dots + m$$

$$y=a+b+c+d+e+f+g+h+i+j+k+l+m \quad (2)$$

**Example 2:** Using eqnarray environment instead of equation environment.

$$y = a + b + c + d + e + f + g + h \nonumber \\ \quad \& \quad \mbox{} + i + j + k + l + m$$

To typeset above, you will get the following output.

$$y = a + b + c + d + e + f + g + h + i + j + k + l + m \quad (3)$$

**Example 3:** Changing the value of `\mathindent` is to change the position that the equation begins<sup>†</sup>.

```
\mathindent=0mm % <-- [A]
\begin{equation}
y=a+b+c+d+e+f+g+h+i+j+k+l+m
\end{equation}
\mathindent=7mm % <-- [B] default value
```

To typeset above (notice [A]), you will get the following output.

$$y = a + b + c + d + e + f + g + h + i + j + k + l + m \quad (4)$$

The value of `\mathindent` must be restored (notice [B]).

### Example 4:

$$\iint_S \left( \frac{\partial V}{\partial x} - \frac{\partial U}{\partial y} \right) dx dy = \oint_C \left( U \frac{dx}{ds} + V \frac{dy}{ds} \right) ds \quad (5)$$

The above equation is too long and almost overrides the equation number. In this case the `\lefteqn` command is useful. It can be used for splitting long formulas across lines as follows.

$$\int_{\partial S} \left( \frac{\partial V}{\partial x} - \frac{\partial U}{\partial y} \right) dx dy$$

---

<sup>†</sup>This explanation is appropriate to left-aligns displayed formulas, not to centering formulas.

$$\oint_C \left( U \frac{dx}{ds} + V \frac{dy}{ds} \right) ds$$

To typeset above, you will get the following output.

$$\begin{aligned} & \int \int_S \left( \frac{\partial V}{\partial x} - \frac{\partial U}{\partial y} \right) dx dy \\ &= \oint_C \left( U \frac{dx}{ds} + V \frac{dy}{ds} \right) ds \end{aligned} \quad (6)$$

**Example 5:** A matrix which is typed by using the array environment.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad (7)$$

The width of a matrix can be shrunk by changing the value of `\arraycolsep` or using an `@`-expression (`@{}`).

```

\begin{equation}
\arraycolsep=3pt %          <--- [C]
A = \left(
  \begin{array}{@{\hskip2pt}%% <-- [D]
             cccc
             @{\hskip2pt}%% <-- [D]
             }
    a_{11} & a_{12} & \& \ldots & a_{1n} \\
    a_{21} & a_{22} & \& \ldots & a_{2n} \\
    \vdots & \vdots & \& \ddots & \vdots \\
    a_{m1} & a_{m2} & \& \ldots & a_{mn}
  \end{array}
\right)
\end{equation}

```

The `\arraycolsep` dimension is half the width of a horizontal space between columns in the `array` environment. A matrix typed by using the `array` environment can be shrunk by changing the value of `\arraycolsep` (notice [C]). And also it can be shrunk by using `@`-expression (notice [D]).

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad (8)$$

Compare Eqs. (7) and (8).

**Example 6:** When you use `pmatrix` `vmatrix` environment etc., same method as Example 5 can be used (`[C]`).

```
\begin{equation}
\arraycolsep=3pt
A = \begin{pmatrix}
a_{11} & a_{12} & \ldots & a_{1n} \\
a_{21} & a_{22} & \ldots & a_{2n}
\end{pmatrix}
```

$$\begin{array}{l}
\vdots \ \& \ \vdots \ \& \ \ddots \ \& \ \vdots \ \backslash \text{cr} \\
a_{\{m1\}} \ \& \ a_{\{m2\}} \ \& \ \ldots \ \& \ a_{\{mn\}} \ \backslash \text{cr} \\
\end{array}
\end{pmatrix}
\end{array}
\end{equation}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad (9)$$

If any of the above explanations could not resolve the problem, there might be the following method, surrounding a display environment with `small` or `footnotesize`, scaling a part or all of a formula by using `\scalebox`, inserting a display environment into a float environment.

## 5. Submission of Editable Electronic Files for Publishing

Information about the submission of editable electronic files for publishing can be given in “Information for Authors (Brief Summary)” and the following web site.  
<https://www.ieice.org/eng/shiori/index.html>

- A source file should constitute a single file. A `bb1` file produced by `BIBTEX` should be include in a main source file.
- Source files required for compilation, such as the original macro file created by authors, special macro files, etc. must be submitted.

## References

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## Appendix A: Printing on A4 paper and making pdf file

- If you print a manuscript on A4 paper by using `dvips` printer driver, the following parameter might be set.

```
dvips -Pprinter -t a4 -O 0mm,8mm file.dvi
```

`printer` is a name of a printer. “`-t a4`” option might be omitted.

- There are three ways to make a pdf file.

1. You can directly make a pdf file by using `pdflatex` as a compiler.

You must specify `pdftex` as an option of `graphicx` package instead of `dvips` etc., and convert from `Figure1.eps` to `Figure1.pdf` by using some tools (for example `epstopdf` etc.).

2. You can convert from a dvi file to a pdf file by using `dvipdfmx`. You must specify `dvipdfmx` as an option of `graphicx` package.

```
dvipdfmx -p 210mm,280mm -x 1in -y 1in
-o file.pdf file.dvi
```

“`-x 1in -y 1in`” option might be omitted.

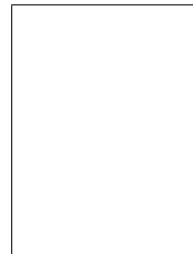
3. You might first convert from a dvi file to a ps file, and then convert from a ps file to pdf file by using Acrobat Distiller. The size of a generating ps file is a4, as there are no definitions of “210mm × 280mm” paper size in `config.ps`.

```
dvips -Pprinter -t a4 -O 0mm,8mm
-o file.ps file.dvi
```

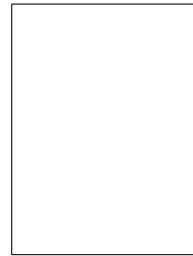
“`-t a4`” option might be omitted.

## Appendix B: Omitted Commands

Some commands which is not required by `ieice.cls` are omitted. These commands are `\tableofcontents`, `\titlepage`, `\part`, `\theindex`, `headings` and the related commands.



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